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ORIGINAL PAPER

Outdoor thermal physiology along human pathways: a study using a wearable measurement system

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Abstract An outdoor summer study on thermal physiology along subjects' pathways was conducted in a Japanese city using a unique wearable measurement system that measures all the relevant thermal variables: ambient temperature, humidity, wind speed (U) and short/long-wave radiation (S and L), along with some physio-psychological parameters: skin temperature (T_{skin}), pulse rate, subjective thermal sensation and state of body motion. U, S and L were measured using a globe anemo-radiometer adapted use with pedestrian subjects. The subjects were 26 healthy Japanese adults (14 males, 12 females) ranging from 23 to 74 years in age. Each subject wore a set of instruments that recorded individual microclimate and physiological responses along a designated pedestrian route that traversed various urban textures. The subjects experienced varying thermal environments that could not be represented by fixed-point routine observational data. S fluctuated significantly reflecting the mixture of sunlit/shade distributions within complex urban morphology. U was generally low within urban canyons due to drag by urban obstacles such as buildings but the subjects' movements enhanced convective heat exchanges with the atmosphere, leading to a drop in T_{skin}. The amount of sweating increased as standard effective temperature (SET*) increased. A clear dependence of sweating on gender and body size was found; males sweated more than females; overweight subjects sweated more than standard/underweight subjects. T_{skin} had a linear relationship with SET* and a similarly clear dependence on gender and body size differences. T_{skin} of the higher-sweating groups was lower than that of the lower-sweating groups, reflecting differences in evaporative cooling by perspiration.

Keywords Mobile observation · Globe thermometer · Outdoor thermal comfort · Outdoor thermal physiology · Mean radiant temperature · Sweat rate

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Introduction

Studies of thermal comfort and thermal physiology generally use fixed-point meteorological data in the target area. This conventional method is useful to capture the overall features of the thermal environment in a given area, but it has two major deficits for investigating the subtle nuances of outdoor thermal comfort and physiology. One is the lack of spatial representativeness of the observed data, and the other is the inability to capture the thermal history of individuals. A point observation can be very different from an observation in the actual environments to which the subjects are exposed when they are moving. Kanda et al. (2006) reported strong heterogeneity of the microclimate and turbulent statistics in urban canyons. Sugawara et al. (2004) mentioned difficulties in obtaining a spatially representative air temperature (T_a) in an urban area. An individual's thermal history can affect their physiological and comfort status at least as much as the current thermal environment. Most previous outdoor comfort studies have directly related the subjective thermal perception/comfort obtained from questionnaires to meteorological factors at just one observation point (e.g. Ahmed 2003; Spagnolo and de Dear 2003; Thorsson et al. 2007a; Pantavou et al. 2013) and thus cannot assess the effect of the thermal history on the dynamic comfort experiences of pedestrians. The

review of the methodology on outdoor thermal comfort experiments was given in Johansson et al. (2014).

Mobile observations can record the spatial and temporal variability of both the microclimate and the physiological states along the pathways of the sensor carriers if the measurement system permits, thereby providing richer microenvironmental information, including thermal histories. Obtaining the overall features of a site requires a composite of numerous samples of mobile observations. The limiting factor for mobile micrometeorological observations is the lack of suitable sensors. Meteorological sensors are designed mostly for stationary observations and are thus not always suitable for mobile applications. Anemometers and short/long-wave radiometers are less portable due to their bulk, weight and power requirements, as well as sensor directionality. Meanwhile, a globe thermometer (GT), which have traditionally been used for brief assessments of human radiant environments (Vennon 1932), are applicable to mobile observations; thanks to their omni-directionality and downsizing potential. A compact GT based on a ping-pong ball has been used to improve response times (e.g. de Dear 1987; Thorsson et al. 2007b). However, GTs still have at least two shortcomings. First, they require the wind speed for the evaluation of mean radiant temperature (MRT), which is the most important meteorological parameter for assessing human energy balance. Second, the radiation properties of GTs can cause under/overestimation of MRT unless they are carefully designed to mimic the human body.

To overcome this problem, Nakayoshi et al. (submitted) developed the "globe anemo-radiometer" (GAR), a unique sensing technology for wind speed (U) and short/long-wave radiation (S and L, respectively). The GAR consists of three compact GTs with different surface properties. Given three globe-based temperatures and concurrent T_a, solving three simultaneous equations of the GTs' heat balance permits the derivation of U, S and L. The GAR evaluates S and L as the mean values of incident radiant energy on the whole globe surface. Thanks to the separate evaluation of the radiation forcing terms, S and L; the GAR can provide an MRT personalised to subjects' radiation properties. The power consumption of the GAR is just 0.2 W. GARs are principally suitable for mobile observations due to their portability, low power consumption and omni-directionality. Additional information on the GAR will be given in "Wearable measurement system" section.

Using the GAR, we developed a unique wearable measurement system (Fig. 1). This instrument measures all parameters required by the main thermal comfort indices along with some important physio-psychological parameters. The observed meteorological variables are T_a, relative humidity (RH), U, S and L. The observed physio-psychological parameters are skin surface temperature (T_{skin}), pulse rate (PR), subjective thermal sensation score (T_{ss}) and state of body motion (BM).

The aim of this study is to propose a new approach in outdoor biometeorological experiment, where subjects observe their individual microclimate and health states along each individual's walking route using the wearable measurement system. A study on outdoor thermal physiology in this approach was conducted with 26 subjects. The study was carried out in Tajimi city, one of the hottest cities in Japan,

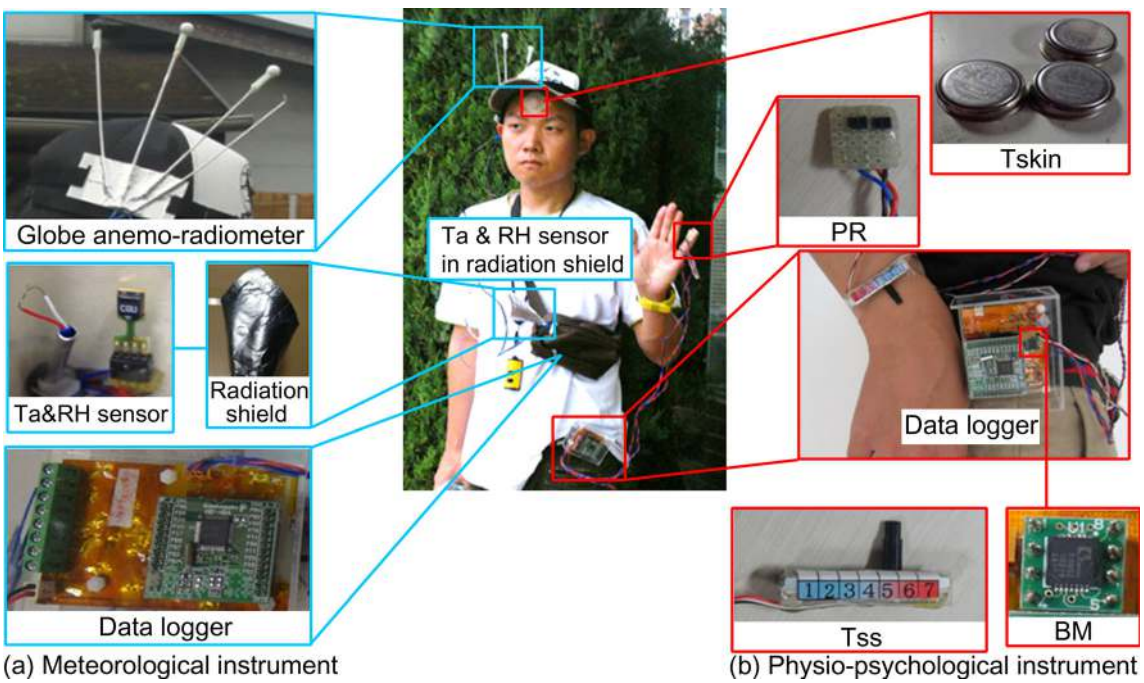


Fig. 1 Wearable measurement system

during summer 2011. In the study, each subject wore our portable measurement system, thus collecting transects of microclimatic and physiological response data along their walking routes. This paper reports the findings from this field study, focusing on the uniqueness of the wearable measurement system.

Wearable measurement system

The wearable measurement system consists of meteorological and physio-psychological subassemblies (Fig. 1).

Meteorological instrument

Figure 2a shows the block diagram of the meteorological instrument. This instrument measures the physiologically relevant thermal environmental variables, T_a, RH, U, S and L, all of which are necessary to evaluate contemporary thermal comfort indices. Each sensor signal is processed and recorded by a miniature data logger. The data logger has four analogue inputs for K-type thermocouples and one digital input for the RH sensor, SHT75 (Sensirion, Inc.). The resolution and accuracy of the analogue inputs in the measurement range of 0–70 °C are 0.02 and 0.2 °C, respectively, while those of the RH sensor are 0.03 and 1.8 %, respectively. One of four analogue input channels was used for T_a measurement with a fine gauge thermocouple (0.05 mm in diameter, Omega Engineering, Inc.). The other three inputs were used for the GAR. The sensors for T_a and RH were enclosed by an aluminium foil radiation shield (Fig. 1).

Globe anemo-radiometer

The GAR consists of three GTs with different surface properties: a black GT (BGT), a white GT (WGT) and, the third, a white GT with internal 0.2-W electrical heat source (WGTH). Provided that the physical properties of a GT, i.e. heat capacity, surface area, albedo, emissivity and heat input, are known a priori, these three globe temperatures combined with T_a can derive heat transfer coefficient (h), S and L by solving three simultaneous equations for GT heat balance:

$$\begin{bmatrix} 1-\alpha_{WGT} & \varepsilon & T_a-T_{WGTH} \\ 1-\alpha_{WGT} & \varepsilon & T_a-T_{WGT} \\ 1-\alpha_{BGT} & \varepsilon & T_a-T_{BGT} \end{bmatrix} \begin{pmatrix} S \\ L \\ h \end{pmatrix} = \begin{pmatrix} \frac{C_{WGTH}}{A}\left(\frac{dT_{WGTH}}{dt}\right) + \varepsilon\sigma T_{WGTH}^4 - H_{input} \\ \frac{C}{A}\left(\frac{dT_{WGT}}{dt}\right) + \varepsilon\sigma T_{WGT}^4 \\ \frac{C}{A}\left(\frac{dT_{BGT}}{dt}\right) + \varepsilon\sigma T_{BGT}^4 \end{pmatrix}, \quad (1)$$

where α_{WGT} and α_{BGT} are albedo for WGT (0.82) and BGT (0.06), respectively, ε is emissivity of GT (0.77), C_{WGTH}/A and C/A are heat capacity of WGTH (733 J K^{-1} m^{-2}) and normal GT (548 J K^{-1} m^{-2}), respectively, σ is the Stefan–Boltzmann constant (5.67×10^{-8} W m^{-2} K^{-4}) and H_{input} is heat input in WGTH (468 W m^{-2}). h is converted to U using an

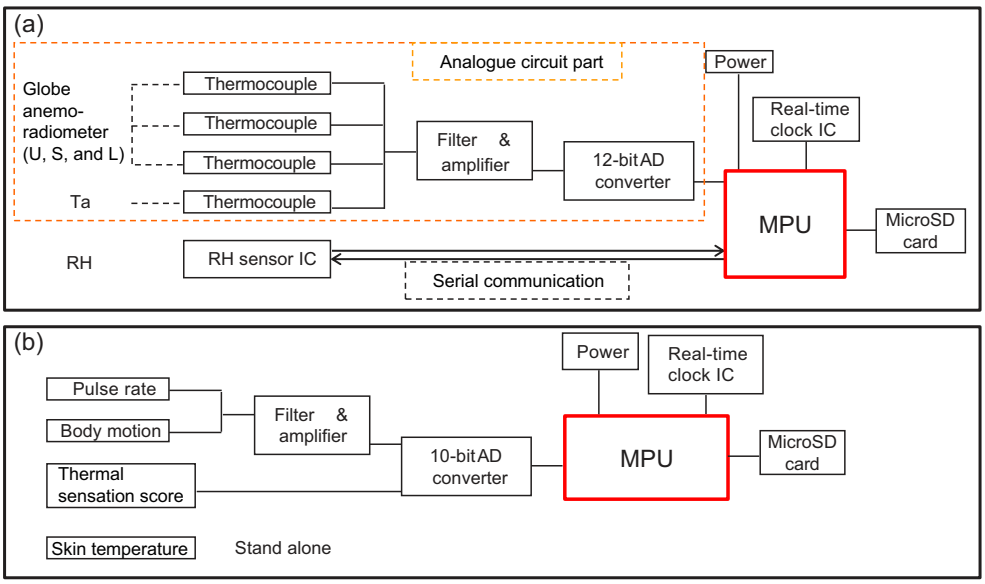


Fig. 2 Block diagram of measurement system: **a** meteorological instrument and **b** physio-psychological instrument

empirical regression. Each GT is much downsized: 12 mm in diameter and 0.3 mm in thickness. Theoretically, the time constant of the GT is approximately 10 s and one tenth of the counterpart of a standard GT (150 mm in diameter), where the time constant is defined as the time required for 63 % response under 1 m s^{-1} wind speed. It should be emphasised here that all of the heat-flux components in Eq. (1) are determined from the unit surface area average of a GT. In practice, complicated urban districts introduce various sources of short/long-wave radiation from different directions. S and L in Eq. (1) do not distinguish these individual contributions but instead include them in a surface average. In stationary use, the accuracies of the GAR for U, S and L with 1 min averaging are 0.24 m s^{-1}, 19.1 and 14.8 W m^{-2}, respectively. All the detailed pieces of information are given in Nakayoshi et al. (submitted).

Configuration of sensors

In wearable use, the body of the subject can adversely affect micrometeorological measurements in three principal ways: the thermal/momentum boundary layer of the body if the sensors are placed there, radiative exchange between the body and the GAR, and BM (especially limbs). These three sources of measurement bias in a wearable system are inevitable but should be minimised through careful design. One compromise design for the GAR was found experimentally: It was installed overhead (Fig. 1) with an aluminium support 150 mm in length and 2.6 mm in diameter. The performance of this layout was evaluated in the following section. T_a and RH sensor enclosed by the radiation shield were placed at least 150 mm apart from the body, and then, the adverse effect by the body was not evident (not shown).

Evaluation of sensor layout of globe anemo-radiometer

The experiment was carried out in the playground of Tokyo Institute of Technology (139° 40′ 49″ E, 35° 36′ 20″ N) during summer 2011. The weather condition was sunny with partial cloud cover. A human subject was instrumented with the GAR on head, as depicted in Fig. 1. The three GTs were exposed approximately 8 cm vertically overhead, with 8-cm horizontal separation between them. Two scenarios were examined: one with subject stepping on the spot next to the reference meteorological sensors and the other with subject walking in a circle around the sensors. The frequency of each step was approximately 1.5 Hz in both scenarios, and the walking speed was approximately 1 m s^{-1}. The experiments were conducted in the morning and around noon to allow an assessment of solar altitude effects. For the reference, two fixed-point observations were performed simultaneously. One used conventional sensors including a four-component net radiometer (MR-40, Eko, Inc.), a pyrheliometer (MS-50, Eko, Inc.) and a three-axis ultrasonic anemometer (CYG81000, R. M. Young Company); the other used a GAR mounted on tripod. The sensors were installed 1.5 m above the ground. Two reference observations were horizontally separated by at least 2 m. From the four-component radiometer and the pyrheliometer, the mean S and L over the sphere's surface were calculated according to the shape factor between the GT and each radiation component:

$$S = sf_{\mathrm{dir}} S_{\mathrm{dir}} + sf_{\mathrm{dif}} \left(S_{\mathrm{dif}} + S_{\mathrm{up}}\right), \tag{2a}$$

$$S_{\mathrm{dif}} = S_{\mathrm{dn}} - S_{\mathrm{dir}} \times \sin(\mathrm{Solar_elevation_angle}), \tag{2b}$$

$$L = sf_{\mathrm{dif}} \left(L_{\mathrm{dn}} + L_{\mathrm{up}}\right), \tag{2c}$$

where S_{dir}, S_{dif}, S_{dn}, S_{up}, L_{dn} and L_{up} are the direct solar radiation, the diffused solar radiation, the global solar radiation, the reflected solar radiation by the ground surface, the downward long-wave radiation and the upward long-wave radiation, respectively; sf_{dir} and sf_{dif} are shape factor for S_{dir} (0.25) and that for the other (0.50), respectively.

Figure 3 shows the time series of U, S and L for the two scenarios conducted at noon. The suffix "_wear" means the GAR in wearable use; "_ref1" and "_ref2" refer to the stationary GAR and the conventional ground-based meteorological sensors, respectively. As shown in the figure, "ref1" and "ref2" were very similar for U, S and L, although the difference for U was larger than that for S and L, probably due to the strong spatiotemporal variability of U. The output of the GAR in wearable use and the other two references showed similar trends in U, S and L, including the larger variation for U. The subject's movement may have contributed to this variation beyond the heterogeneity of U. Because the GAR uses U to calculate S and L, a match in both S and L indicates that "U_wear" correctly senses the actual U around the GTs considering the subject's movement. A similar result was obtained in the morning session (not shown). Thanks to the similarity of the experimental conditions, i.e. season, climate and clothing and skin conditions of subjects; between this preexamination and the experiment in the real city, data observed by the wearable measurement system in the target city can represent the microclimate there without significant effects of the human body. For robust testing of the suitability of the wearable instrument for field studies of thermal comfort in different terrain and texture types, seasons and times of day, additional studies are necessary.

Physio-psychological instrument

The physio-psychological parameters, T_{skin}, PR, T_{ss} and BM were measured. Each sensor signal is processed and recorded by a generic data logger, with the exception of T_{skin} (Fig. 2b).

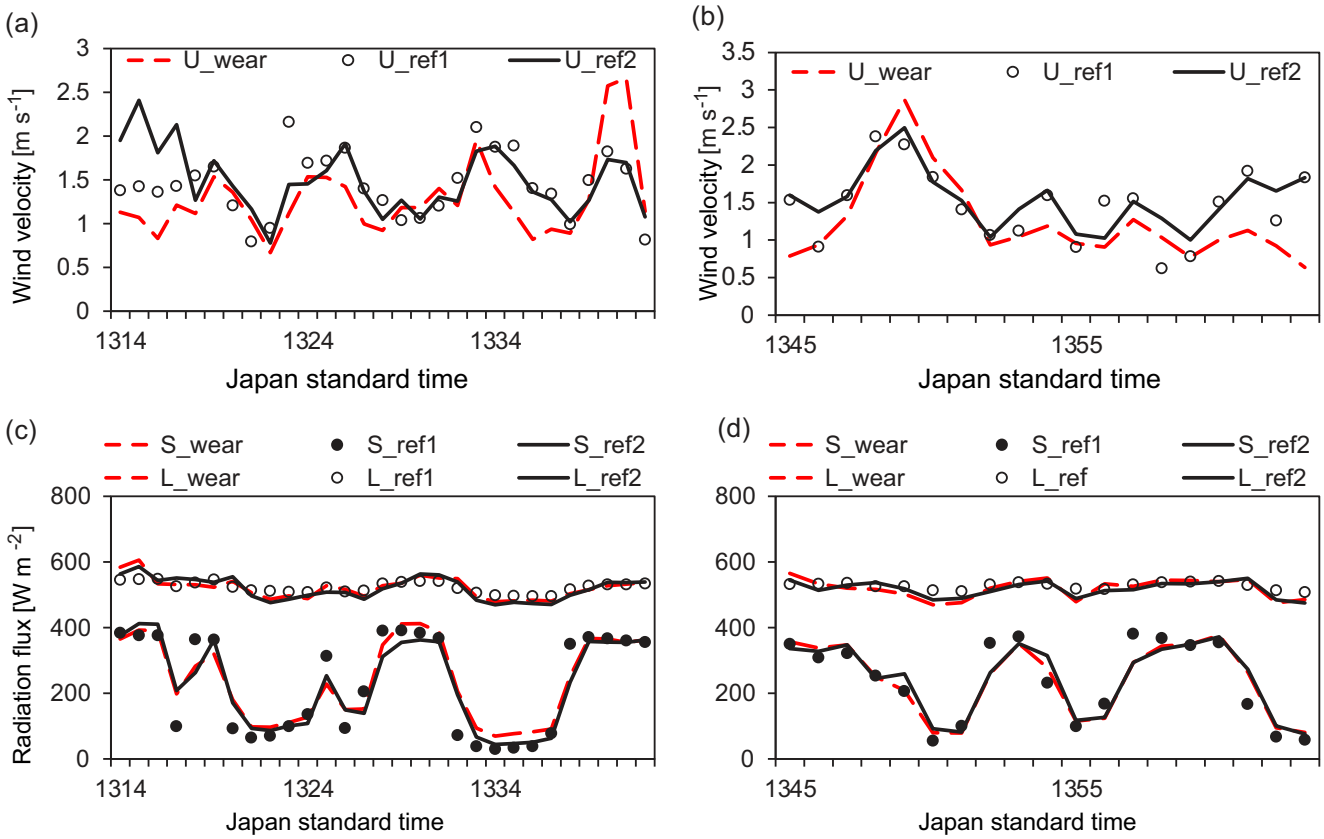


Fig. 3 Accuracy of the wearable globe anemo-radiometer. Wind speed when standing on the spot (**a**) and walking around the reference sensors (**b**). Short/long-wave radiation when standing on the spot (**c**) and walking around the reference sensors (**d**). The suffix "*_wear*", "*_ref1*" and "*_ref2*" mean GAR in wearable use, GAR in stationary use and conventional sensors, respectively

T_{skin} is measured using the Thermochron iButton (SL type, KN Laboratories, Inc.), a compact stand-alone data logger for temperature. This sensor has been used for body temperature measurement in animal biometeorological field studies (e.g. Lovegrove 2009). Each Thermochron iButton is attached to the skin surface and covered with thermal insulation material (surgical tape) to exclude the effect of T_a. To assess the mean T_{skin} over the entire body surface, the T_{skin} of seven different body parts (forehead, belly, left forearm, left hand, left thigh, left lower thigh and left ankle) was measured following Hardy and DuBois (1938). Hereafter, T_{skin} is taken to refer to the average value of these seven skin sites. PR is measured by a customised sensor consisting of an infrared light-emitting/receiving diode that detects the volume pulse wave of a finger. This kind of PR sensors has been extensively used and well validated in sports medicine research. An adjustable resistor is used to register thermal sensation (T_{ss}). The sensor was marked with seven levels according to the ASHRAE guidelines on subjective thermal sensations: 1=cold, 2=cool, 3=slightly cool, 4=neutral, 5=slightly warm, 6=warm and 7=hot. Body motion (BM) is measured using a tri-axial accelerometer. Frequency analysis of the accelerometer signals provides BM, as shown in Fig. 5h.

Methods

Experimental design

The experiment was carried out in Tajimi city (137° 7′ 12″ E, 35° 20′ 5″ N), one of the hottest cities in Japan, during August 22–24, 2011. The subjects were 14 healthy males and 12 healthy females ranging from 23 to 74 years in age. Each subject gave their informed consent in writing prior to the experiment. Table 1 shows the subjects' anthropometric characteristics. Three observations were held each day, in the morning, at noon and at sunset, with subjects in different age brackets: those in their 30s and 40s were observed on Aug. 22, those in their 20s on Aug. 23 and those 50 and over on Aug. 24. All subjects wore the instrumentation described above to monitor their personal microclimate and physiological responses along the observational route (Fig. 4). The route traversed various urban textures including a parking lot (Fig. 4, Ia), commercial areas (Fig. 4, Ib), an area with reflective paint (Fig. 4, Ic), biotope areas (Fig. 4, Id) and green spots (Fig. 4, Ie). In addition to the meteorological and physio-psychological variables mentioned above, the amount of sweat loss throughout the experiment was also measured using precision digital scales HJR-62 K and CG-100KF

Table 1 Subject information and route of observations

ID	Gender	Age	BSA (m^2)	BMI	Route
A1	Male	47	1.79	19.14	Ia–Ia–Ib–Ic–Id–Ie
A2	Male	35	1.59	22.27	Ia–Ib–Ic–Id–Ie–Ia
A3	Male	45	1.99	29.07	Ia–Ic–Id–Ie–Ia–Ib
A4	Male	39	1.97	27.73	Ia–*Id–Ie–Ia–Ib–Ic
A6	Female	41	1.42	18.03	Same as A1
A7	Female	30	1.57	19.47	Same as A2
A8	Female	45	1.34	19.11	Same as A3
A9	Female	44	1.55	20.45	Same as A4
A10	Female	35	1.45	18.83	Ia–*Ie–Ia–Ib–Ic–Id
B1	Male	24	1.76	22.23	Ia–Ib–Ic–Id–Ie
B2	Male	24	1.76	20.81	Ia–*Id–Ie–Ib–Ic
B3	Male	24	1.69	21.61	Ia–Ic–Id–Ie–Ib
B4	Male	23	2.01	25.59	Same as B3
B5	Male	23	1.56	19.95	Ia–*Ie–Ib–Ic–Id
B6	Male	25	2.16	28.07	Same as B5
B7	Female	23	1.60	20.82	Same as B1
B8	Female	23	1.44	20.93	Same as B2
C1	Male	64	1.88	24.73	Ia–Ia–Ib–Ic–Id–Ie
C2	Male	74	1.46	19.23	Ia–Ib–Ic–Id–Ie–Ia
C3	Male	68	1.77	23.46	Ia–Ic–Id–Ie–Ia–Ib
C4	Male	55	1.80	22.39	Ia–*Id–Ie–Ia–Ib–Ic
C6	Female	62	1.43	20.24	Same as C1
C7	Female	57	1.73	23.31	Same as C2
C8	Female	53	1.49	21.64	Same as C3
C9	Female	54	1.44	18.20	Same as C4
C10	Female	69	1.45	20.40	Ia–*Ie–Ia–Ib–Ic–Id

In the route column, underbars indicate the performance of *Uchi-mizu* before the 5-min rest. Asterisk followed by dash (–*) represents the transport of subjects by bus

BSA mean body surface area, *BMI* mean body mass index

(Shinko Denshi Co., Ltd.). Their accuracies are 1 g for the HJR-62 K and 5 g for the CG-100KF, respectively. The amount of sweating was assessed as the difference between the body weight and weight of a water flask before and after the experiment, as the subjects were asked not to discharge urine during the experiment.

The measurement procedures in each experimental session were as follows:

1. Before the session, every subject measured his/her naked weight and the weight of their water flask full of water. Then, they put on the measurement system.
2. All subjects were taken to point Ia (Fig. 4) by bus.
3. The subjects formed four or five groups, each with a spotter. Spotters sprinkled water around the subjects to assess the effect of *Uchi-mizu* (explained later) on their thermal perceptions. They remained there for 5 min in a natural standing posture.
4. Each group then moved to the next points noted in Table 1. The groups whose destinations were point Id or Ie were taken there by bus.
5. After all groups arrived at their destinations, all subjects simultaneously started a 5-min rest period in a standing posture.
6. After the rest, each group walked to the next point following a clockwise pattern. They then repeated step 5.
7. When the subjects had visited all the locations, the experimental session ended. They then returned by bus to the anteroom where their naked weight and the weight of the water flask were re-measured.
8. All subjects waited until the next experiment inside an acclimation chamber with a thermally neutral environment, where room temperature was set to 28 °C with dehumidification.

Uchi-mizu is a traditional Japanese method of mitigating excessive environmental heat in summer. Water is sprinkled on urban surfaces (streets, roads, gardens etc.), and the latent heat of vaporisation cools surfaces, reducing sensible and radiant transfers onto the occupants.

Thermal comfort index

Standard effective temperature (SET*, Gagge et al. 1971) was selected from the dozens of composite comfort indices available in the literature. SET* is a steady-state model based on the Pierce Lab's two-node model for human energy balance, and with over 40 years of service, it represents an optimal balance between physiological complexity and computational/operational simplicity. The model is referred to as "two-node" because it simulates heat exchanges between body core (inner node) and peripheral tissue (outer node) through various thermoregulatory processes such as vasomotion, heart rate and metabolism. It also calculates heat exchanges between skin surface and surrounding microclimate. SET* relies on the strong correlation between skin wettedness (fraction of skin wet with sweat) and subjective discomfort in heat/humidity. In cold environments, skin temperature is the main predictor for subjective discomfort in SET*. Being a steady-state model SET* is theoretically inapplicable to transient situations such as the scenarios studied in this project as is the case for most thermal comfort indices in current use. Therefore, the use of SET* was limited to an assessment of the integrated thermal microclimate at each point along the observational route, assuming that the subjects reached thermal steady-state conditions at each point. Particularly, 60-min physiological iterations were applied in the two-node model, with 1-min averaged thermal data at each point. As for the non-environmental input parameters of clothing insulation



Fig. 4 Observational route and photos of each location where 5-min rest periods occurred

and metabolic rate, the same values were applied to all subjects: 0.5 clo in thermal insulation (average value of the sample of subjects), 0.3 in albedo α (average across the sample of subjects), 0.97 in emissivity ε (typical value of the human body), 2.0 Met units (typical metabolic value for walking) and 1.72 m^2 in body surface area (typical value for young Japanese subjects). Every subject wore a yellow vest over their own clothes to standardise body surface radiation properties. Under the above scenario, MRT, an important meteorological input for SET*, is evaluated as follows:

$$\mathrm{MRT} = \sqrt[4]{\frac{(1-\alpha)S + \varepsilon L}{\varepsilon\sigma}}. \quad (3)$$

There are three major methods to evaluate MRT: six-directional short- and long-wave radiation method (Verein Deutscher Ingenieure: VDI, 1998), standard GT method (ISO 1998), and method with direct, diffuse and reflected solar radiation and upward and downward long-wave radiation (Spagnolo and de Dear 2003). MRT derived by VDI method is essentially different with that by our method because of no consideration of direct solar radiation in VDI (1998). MRT by our method can be convertible with MRT by ISO (1998) given the radiation properties of the GT. MRT by Spagnolo and de Dear (2003) can be also comparable with that by our method because the similar manner were taken in the validation of the GAR (see "Evaluation of sensor layout of globe anemo-radiometer" section).

Result and discussion

Meteorological and physiological variations along human pathways

Figure 5 shows the time series of meteorological and physiological data along the pathways of subjects C4 (male) and C9 (female). Stationary meteorological data observed by the Japan meteorological agency were also used as a reference: global solar radiation (S_{Global}) at 136° 57.9′ E and 35° 10′ N and T_a, humidity and U at 136° 45.7′ E, 35° 24 N. The reference T_a and humidity were measured at 2 m above the ground, and the reference U was measured at 23 m above the ground. The weather condition during this experimental session was fine with partial clouds. Because the subjects walked together, the microclimate observed by C9 was the almost the same as that for C4 and is thus not shown. The grey-shaded box indicates that the subjects were transported by bus.

Similarity between SET* (Fig. 5a) and MRT (Fig. 5b) traces suggests that radiant load on the subjects was the main thermophysiological variable in these scenarios. Shortwave flux, S (Fig. 5c), was more influential on MRT variations than was long wave, L (Fig. 5d). The high fluctuation in S compared with S_{Global} reasonably reflected the mixture of sunlit/shade distribution within the complicated urban districts. For instance, the biotope (Id in Fig. 5c) and the urban green space (Ie in Fig. 5c) had lower S despite the high S_{Global}, probably resulting from the shading effect of tree crowns.

T_a and T_a_reference (Fig. 5e) gradually increased throughout each experimental exposure due to diurnal environmental cycles. The higher variation in T_a compared with T_a_reference reflected differences in urban textures and morphology. The biotope (Id) and the urban green space (Ie) had lower T_a than T_a_reference. As expected, the vegetation and the waterside formed cool spots. The sharp drop in T_a around 10:00 was due to "water mist spraying (WMS)", open-air evaporative cooling equipment developed for heat mitigation in urban in Japan (Ishii et al. 2009).

In contrast to T_a, no clear diurnal trend in specific humidity (q) was seen in either the mobile (q) or reference data (q_reference) (Fig. 5f). A spike in q around 10:00 coincides with a sharp drop in T_a due to WMS. The spatial variability of q will be discussed in statistical terms in the composite analysis.

Air speeds recorded by the human subjects, U (Fig. 5g), were generally lower compared to stationary reference observations. This is probably due to the drag of urban obstacles such as buildings and trees. One noticeable feature is that higher U corresponded to the walking periods (identified by high BM in Fig. 5h). This suggests that walking subjects' movements enhanced their convective and evaporative heat loss, causing a drop in T_{skin} (Fig. 5j).

Pulse rate (PR; Fig. 5i) correlates well with body movement, BM; subjects who were walking had elevated BM and corresponding increased PR. The clear relationship between PR and the thermal variables observed in indoor experiments (e.g. Niimi et al. 1997) was not evident in these outdoor studies both in this transient analyses and composite analyses (not shown). The different mechanism on PR increase between heat stress and exercise may perplex the PR behaviour under the both influences. The increase of PR by exercise is related to energy generation; Bernard et al. (1997) reported the relation between PR and oxygen intake during exercise. Meanwhile, the enhancement of PR under heat stress will occur to release the heat efficiently from the internal body.

Fig. 5 Time series charts of the output of the wearable measurement system (subjects C4 and C9). The *grey-shaded box* represents transport of the subjects by bus. **a** SET*, **b** MRT, **c** S and S_{Global}, **d** L, **e** T_a, **f** q, **g** U and dominant wind direction, **h** BM, **i** PR, **j** T_{skin} and **k** T_{ss}

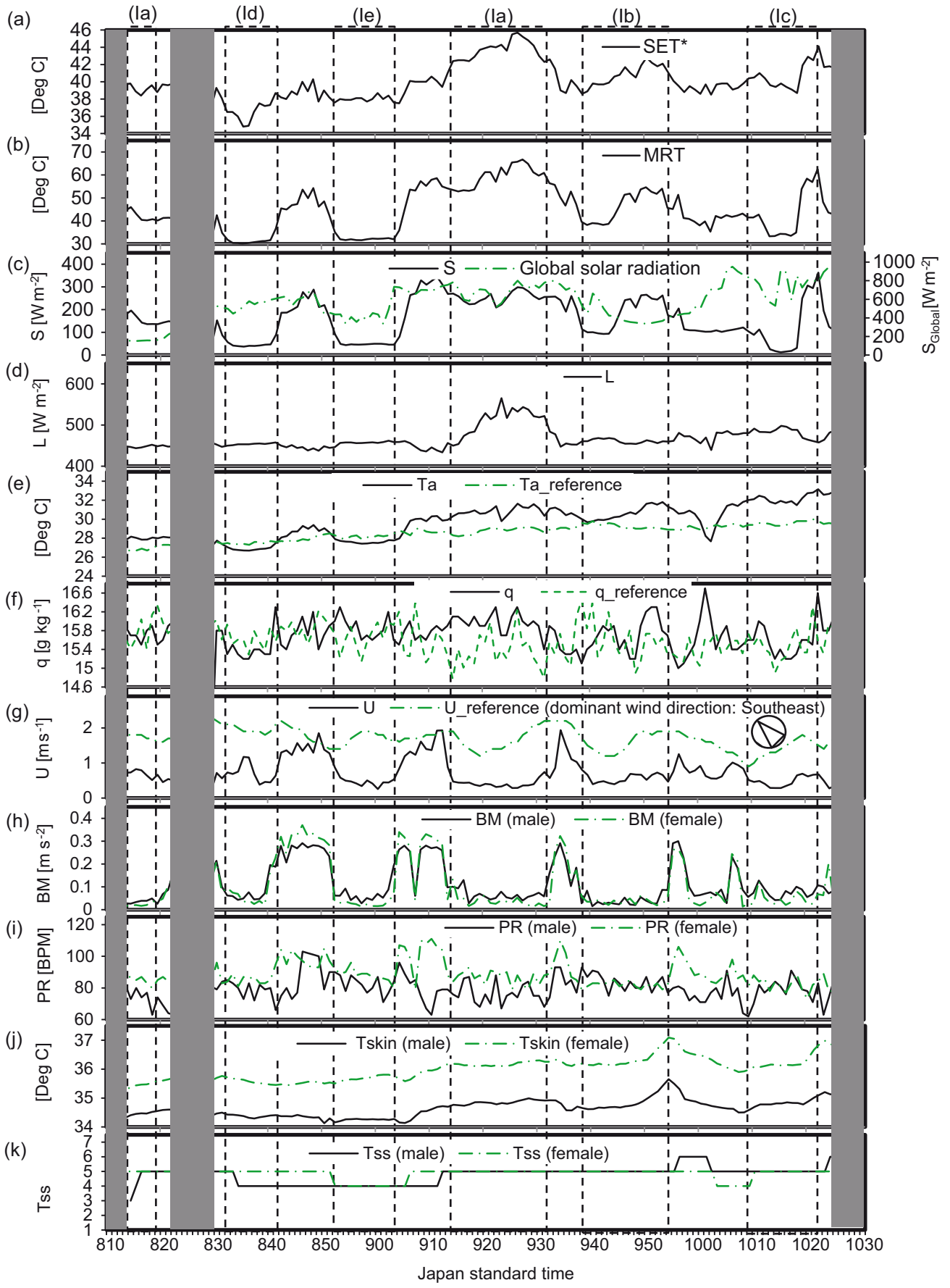
T_{skin} (Fig. 5j) gradually increased with elapsed time, corresponding to the warming of atmosphere, T_a. The fluctuation of T_{skin} throughout this upward trend corresponded to variations in SET*. The relationship between T_{skin} and SET* is evidenced by the statistical analysis in Fig. 9. However, the frequency of T_{skin} fluctuations was slower than that of SET*, reflecting the thermal inertia of body tissue and also a feedback effect from thermoregulatory functioning. Among the environmental inputs to SET*, S had the greatest influence on T_{skin} in the morning and midday sessions. The increases in T_{skin} around 9:04, 9:55 and 10:20 coincided with increases in S. The effect of walking-induced U on T_{skin}, as discussed above, became more evident in the sunset session when S was negligible (Fig. 6). The enhanced U (Fig. 6a) due to the subjects' movements (Fig. 6b) effectively cooled T_{skin} (Fig. 6c). Differences in T_{skin} between the male and female subjects will be discussed later.

T_{ss} (Fig. 5k) varied around "slightly warm", with short-lived excursions to "neutral" and "warm". The values of T_{ss} were likely related to those of T_{skin}, but with a certain lag time. One potential explanation for this lag time was an experiment procedural artefact; subjects were not required to note T_{ss} at certain intervals. Another possible reason is the difference in the physical processes underlying T_{skin} and T_{ss}. Because skin is the boundary between atmosphere and the body, T_{skin} had a relatively direct response to the changes in thermal environmental conditions. However, the central nervous system plays an important role in regulation of T_{ss} (de Dear 2011), contributing further lags in the relationship between skin and environmental temperatures.

The above discussions essentially apply to the other data, despite differences in subjects and time frames.

Composite analyses

Preceding discussions have deliberately ignored differences between subjects and also spatiotemporal dynamics in microclimate, but from the human perceptual point of view, these


(a)
(Ia)
(Id)
(Ie)
(Ia)
(Ib)
(Ic)
SET*
[Deg C]
(b)
MRT
[Deg C]
(c)
S
Global solar radiation
S [W m-2]
SGlobal [W m-2]
(d)
L
L [W m-2]
(e)
Ta
Ta_reference
[Deg C]
(f)
q
q_reference
q [g kg-1]
(g)
U
U_reference (dominant wind direction: Southeast)
U [ms-1]
(h)
BM (male)
BM (female)
BM [m s-2]
(i)
PR (male)
PR (female)
PR [BPM]
(j)
Tskin (male)
Tskin (female)
[Deg C]
(k)
Tss (male)
Tss (female)
Tss
810 820 830 840 850 900 910 920 930 940 950 1000 1010 1020 1030
Japan standard time

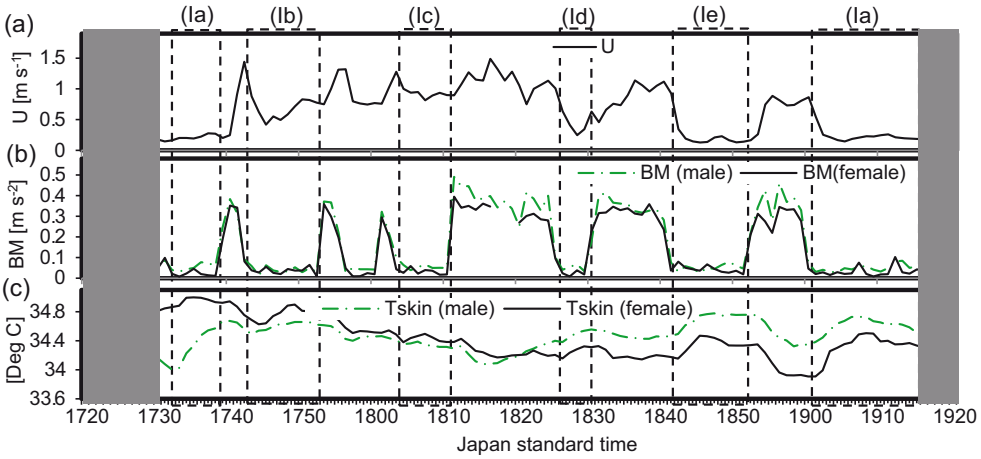


Fig. 6 Time series charts of the output of the wearable measurement system (subjects A2 and A7). The *grey-shaded box* represents transport of the subjects by bus

have a strong impact on experience of the thermal environment.

Spatial variations of microclimate

We examined the spatial characteristics of microclimates by comparing the parking lot (Ia), the commercial area (Ib), the reflective paint area (Ic), the biotope (Id) and the green spot (Ie). Because the raw data included spatiotemporal information, the following procedures were taken to remove temporal feature. First, the anomalies from the spatial average in the five locations where simultaneous measurements were recorded by different subjects were identified. When locations had missing data, the spatial average was assessed from the other data. Spatial

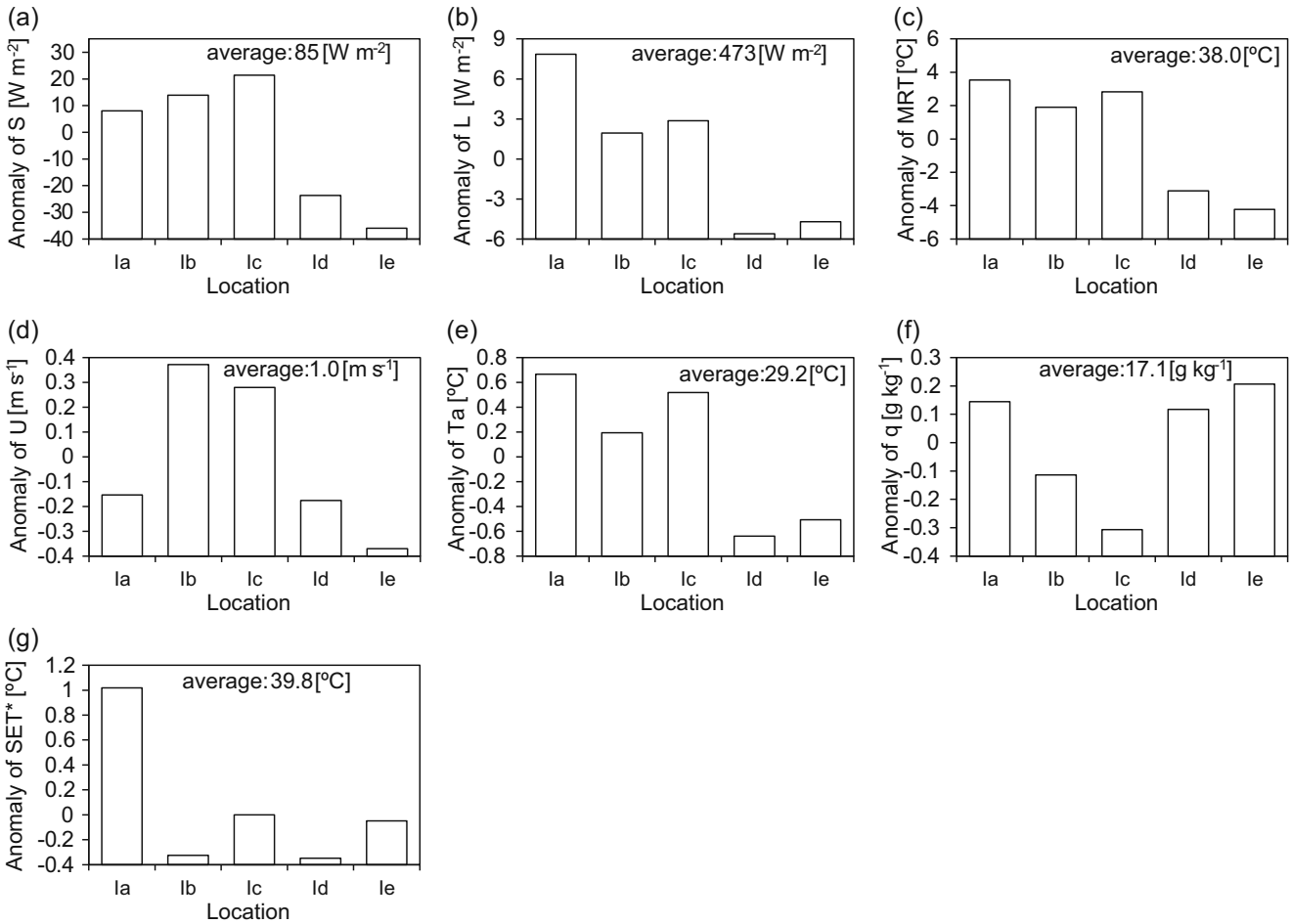


Fig. 7 Spatial anomaly of the thermal variables: **a** *S*, **b** *L*, **c** MRT, **d** *U*, **e** T_a, **f** *q* and **g** SET*

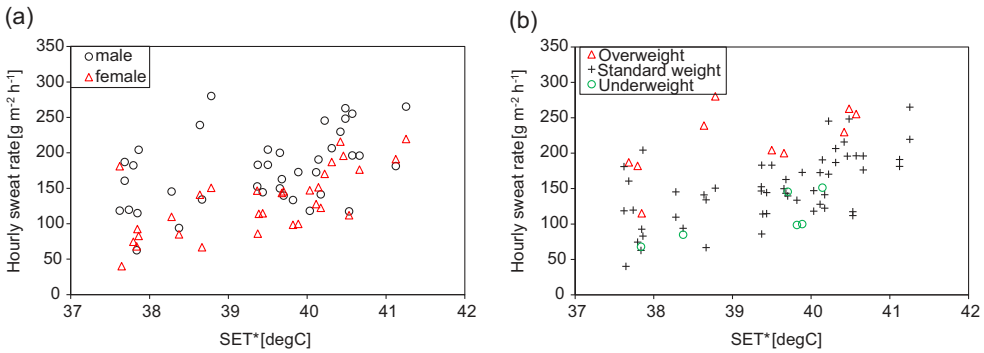


Fig. 8 Relationship between the hourly sweat rate per unit skin surface area and SET*: difference by **a** gender and **b** by body size

anomalies were then temporally averaged. A total of 210 min of data, including morning, noon and sunset times, were averaged.

S (Fig. 7a) was lower in the vegetated areas, Id and Ie, compared to the urban areas, Ia, Ib and Ic, as a result of shading by tree crowns. Higher albedo at Ic probably explains the higher *S* value compared to the other urban areas (Ia and Ib). The relatively small *L* (Fig. 7b) in the vegetation areas can be explained by the lower surface temperature resulting from evaporative cooling by vegetation. Consequently, MRT (Fig. 7c) was higher in the urban areas than in the vegetation areas.

Flow along a major river south of the experimental area (Fig. 4) possibly contributed to higher *U* values (Fig. 7d) at Ib and Ic. These areas were located on a relatively wide street running parallel to the river, and the wind direction during the study was often along the river.

Air temperature, T_a (Fig. 7e), was highest at Ia, followed by Ib, Ic, Ie and Id, in that order. Higher MRT (Fig. 7c) possibly contributed to higher T_a values in the urban areas. Additionally, the reduction in heat diffusion and advection by the lower wind speeds *U* (Fig. 7d) at Ia compared with the other urban locations could have caused the higher T_a. In addition to the smaller MRT in the vegetated area, the evaporative cooling by the greenery and the water body also contributed to the lower T_a there, evidently leading to a higher value of *q* (Fig. 7f). The higher *q* at Ia can be explained by the *Uchi-mizu* performed at the beginning of each experimental session and the smaller *U*, which limited the diffusion/transport of *q*.

Integrating all of these microclimatic variations, SET* peaked at Ia, largely due to elevated MRT, T_a and *q*, combined with lower air speeds, *U*; this was followed by Ic, Ie, Ib and Id, in that order. The smaller SET* values registered in the vegetated areas were as expected. Unexpected, however, were the lower SET* values recorded at Ib and Ic, despite their urban textures. This counter-intuitive finding can probably be attributed to elevated wind speed, *U*, caused by drainage flows along the adjacent river valley.

Thermal environmental effects on human physiology

The correspondence of thermophysiological processes and the SET* as a comprehensive human thermal discomfort index can be examined. Figure 8 shows the relationship between the observed sweat rates and SET*. The sweat rate was normalised to the body surface area of each subject and averaged over the duration of the study. SET* was also averaged over the experimental exposure. As might be expected, sweat rates increased with increasing SET*. Sweat rate was also clearly related to both gender and body size. Here, the body size is assessed by the body mass index (BMI). Males had higher sweat rates than females, which corresponds to findings of indoor experiments (e.g. Fox et al. 1969). The gender difference is explained by the higher sweating capacity of males. The relationship between BMI and sweat rate matches everyday observations of obese persons sweating more heavily under an imposed environmental heat load than more slightly built counterparts. Fat tissue represents a thermal insulation layer between the blood capillaries (heat exchangers) and the atmosphere surrounding the body, reducing sensible heat

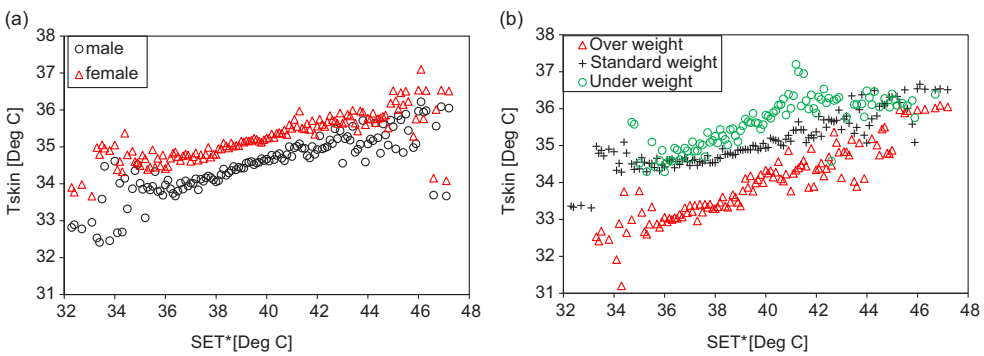


Fig. 9 Relationship between the average skin temperature and SET*: difference by **a** gender and **b** by body size

conduction to skin surface, causing heat accumulation in obese subjects and relatively higher rates of sweating compared to ectomorphic and mesomorphic subjects.

Figure 9 shows the relationship between T_{skin} and SET*. The 1-min averaged T_{skin} of all the subjects was further averaged for every 0.1 K of SET*, taking gender and body size differences into account. A clear dependence of T_{skin} on SET* was revealed by this analysis. Additionally, gender and BMI differences were observed once again. The more heavily sweating male and overweight subjects had lower T_{skin} than did their lighter, female counterparts, but the T_{skin} differences diminished as SET* increased. The biases in T_{skin} could have been caused by differences in latent heat vaporisation due to different perspiration capabilities, between males and females and also between overweight and lighter subjects. The different behaviour of T_{skin} biases between low SET* and high SET* is explicable in terms of the wettedness parameter of the skin surface, i.e. the fraction of the body's skin surface area wet due to sweating over the whole area. As SET* increased, skin wettedness parameter of the subjects could approach unity, and between-subject differences in evaporative cooling diminish accordingly. In low SET* environments, the subjects' skin wettedness is far from 100 %; therefore, T_{skin} more clearly differentiated subjects according to gender and BMI.

Concluding remarks

Our experience of the thermal microclimate in outdoor settings is vastly more complicated compared to the indoor context, largely because of the non-steady-state nature of the exposure. Most of the time we spend in urban outdoor settings is on the move, and the acute dynamic sensitivity of our thermal perceptual systems, particularly our cutaneous thermoreceptors, heightens our awareness of even the most subtle urban microclimatic nuances (de Dear 2011), both as we traverse them and as they change themselves from one moment to the next. These observations render conventional approaches to thermal comfort measurement with stationary instrumentation and simulation steady-state comfort models inappropriate (Jendritzky et al. 2012).

This outdoor study on thermal physiology and comfort was conducted in Tajimi city, one of the hottest Japanese cities, in summer 2011 using a unique wearable measurement system. A unique wearable measurement system consisting of various micrometeorological and physio-psychological sensors was developed to more faithfully represent the dynamics of the subjects' exposure to the complex patchwork of urban microclimates they traversed. The accuracy of the meteorological instruments in wearable use was carefully validated against standard meteorological reference instruments. Each subject wore their own measurement system. Individual microclimate and physiological responses along the observational route covering various urban textures were then assessed.

The subjects experienced wide variations in SET* within the complicated urban district. In the morning and midday experimental sessions, widely fluctuating S was recorded in response to the mixture of sunlit/shade distribution by urban obstacles and tree crowns. These dynamics in the radiant environment of the pedestrians were the dominant contributor to the highly variable SET* conditions experienced during their Lagrangian exposure. Urban greenery and biotope represented discernible cool spots along the traverse. Within the urban canyon, U was generally low but the subjects' BM enhanced convective heat exchange with the atmosphere, causing T_{skin} to drop. As expected, the amount of sweating increased as SET* increased. Moreover, a clear dependence of the amount of sweating on gender and body size difference was found; males sweated more profusely than the females in our sample, and overweight subjects sweated more than standard and underweight subjects. T_{skin} showed a linear relationship to SET* but also had a clear dependence on gender and body size difference. Subjects with higher levels of sweating registered lower T_{skin}, reflecting a difference in evaporative cooling by sweat. These differences were extinguished as SET* increased due to the wettedness parameter approaching unity.

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